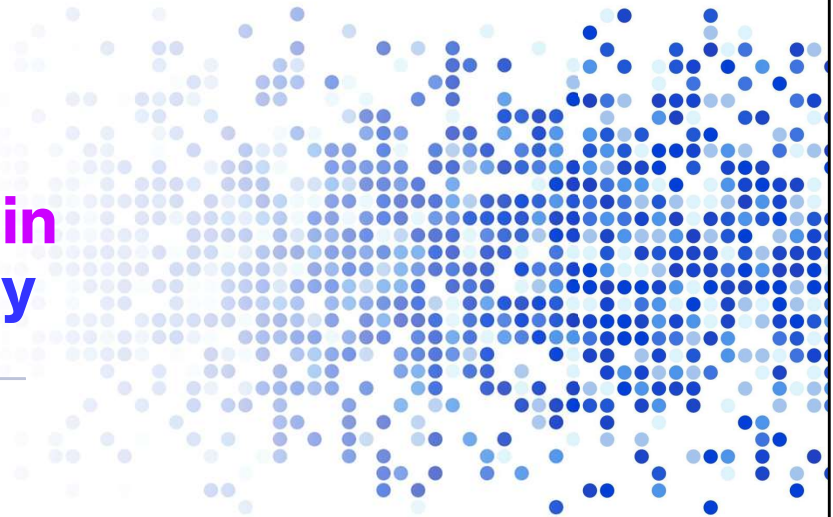




Prostaglandin Biochemistry

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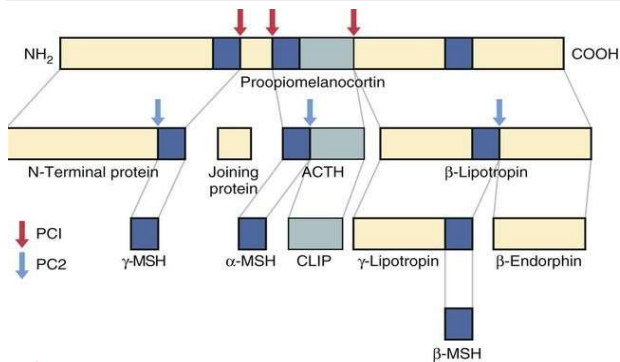


Learning Objectives

1. Describe the structures of fatty acid precursors of prostaglandins, and their dietary and metabolic sources.
2. Identify cyclooxygenases and lipoxygenases as key enzymes in eicosanoid metabolism.
3. Distinguish between COX-1 and COX-2.
4. List the most important products of cyclooxygenase and lipoxygenase pathways, as well as their general structural features.
5. Explain the meaning of the subscript numbers of prostaglandins.
6. List the most important actions of prostaglandins.
7. Explain how prostaglandin synthesis can be inhibited

Biochem Express 1!!!

Proopiomelanocortin (POMC)



Synthesized primarily in the **anterior pituitary**

ACTH (corticotropin)– secretion signal of the glucocorticoids

β-lipotropin- ↑lipolysis (adipocytes)

γ-lipotropin- ↑FA release

CLIP- ↑insulin release

β-endorphin – endogenous opioid; more potent than morphine; ↓pain perception

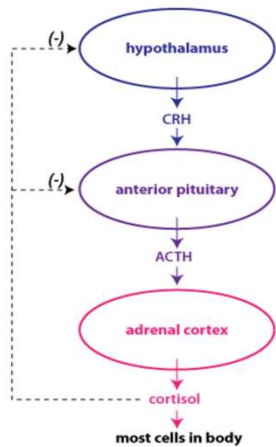
α-MSH - ↑melanin synthesis

β-MSH - ↓hunger ↑satiety ↑melanin

γ-MSH - ↑Na⁺ excretion ↓BP

Biochem Express 2!!!

Regulation of Cortisol Secretion



CRH: Corticotropin releasing hormone

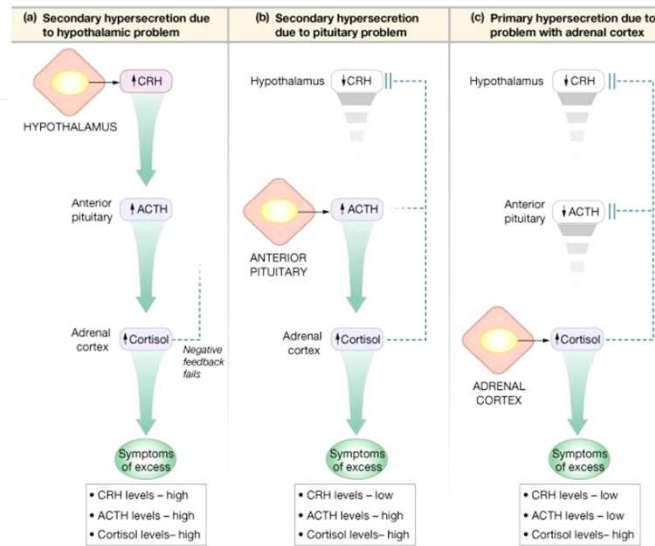
ACTH: Adrenocorticotrophic hormone (corticotropin)

ACTH exerts its effect via **cAMP**.

Clinical Point: **CUSHING SYNDROME**

1. Hypercortisolism
2. Causes: Iatrogenic (exogenous), **tumor of the pituitary** (**↑↑ ACTH, Cushing disease**), tumor of the adrenal gland
3. Tests: 24-hour urinary free cortisol test
Dexamethasone suppression test
4. **Buffalo hump, moon-faced, hirsutism and abnormal menses (females)**, central body obesity, ecchymosis, muscle atrophy, depression, hypernatremia, hypokalemia, hypervolemia, hypertension, hyperglycemia, immunosuppression

Biochem Express 3!!!



Outline

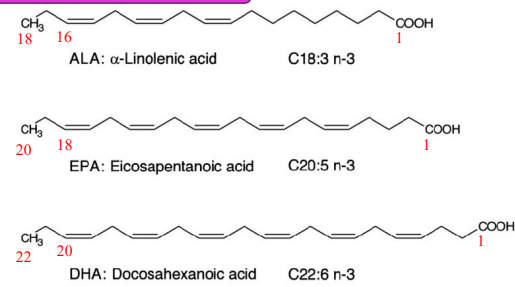
1. Omega-3 and 6 Fatty Acids
2. Substrates for Prostaglandin Synthesis
3. Eicosanoid Synthesis
4. Cyclization of Arachidonic Acid: Cyclooxygenase Path
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6. Omega-3 FA and Eicosanoid-related Compounds
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Omega-3 and 6 Fatty Acids

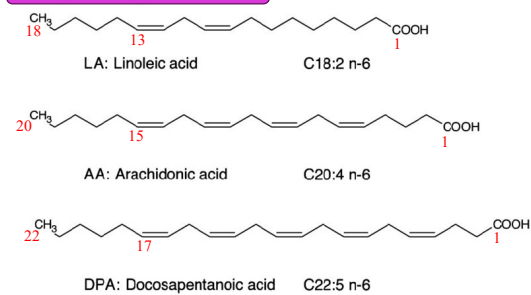
• Essential Fatty acids

- PUFA that must be taken exogenously
- **Linoleic acid** (18:2^{Δ9,12}) is the precursor of other ω-6 fatty acids like arachidonic acid (AA, 20:4) and DPA (22:5) (plant oil)
- **Linolenic acid** (18:3^{Δ9,12,15}) is the precursor of other ω-3 fatty acids like **EPA** (20:5) and **DHA** (22:6) (fish oil, nuts, seeds)

Omega-3-Fatty Acids



Omega-6-Fatty Acids



Structure of **α-linolenic acid**. EPA is an n – 3 PUFA derived from **fish** oil. It is a **precursor** of **prostaglandin E₃** and some other eicosanoids. The fats from marine animals have greater quantities of longer carbon chain fatty **acids** and with higher levels of unsaturation when compared to mammalian fats.

Linoleic acid and linolenic acid are termed essential fatty acids because animals cannot synthesize them and must acquire them in their diet

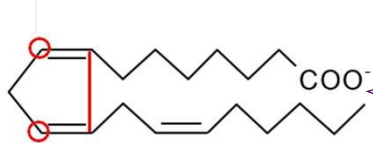
Linoleic acid is the precursor of arachidonic acid, and both of these molecules are termed ω6 fatty acids

ω6 fatty acids are precursors of prostaglandins, thromboxanes, and leukotrienes – inflammatory local hormones

Linolenic acid is the precursor of eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), and these three molecules are termed ω3 fatty acids (n-3)

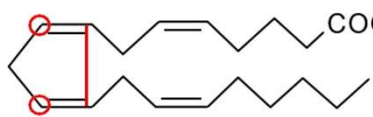
ω3 fatty acids are cardioprotective, anti-inflammatory, and anticarcinogenic

Substrates for Prostaglandin Synthesis



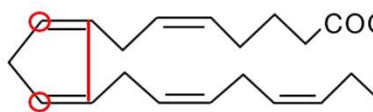
Eicosatrienoic acid (20:3)

- Omega-6 FA
- Makes PGE_1
- Found in nuts, plant oils and seeds



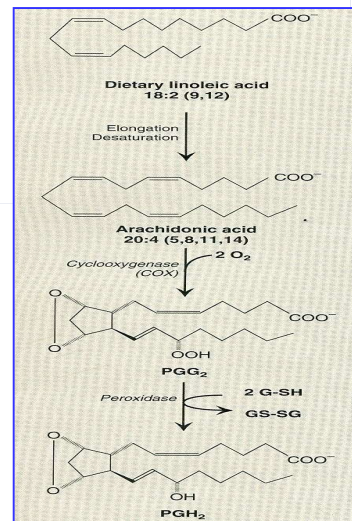
Arachidonic acid (20:4)

- Omega-6 FA
- Makes PGE_2
- Found in nuts, plant oils and seeds
- Most important substrate because of its abundance



Eicosapentaenoic acid (EPA, 20:5)

- Omega-3 FA
- Makes PGE_3
- Found in fish oils and algal oils



Conversion of 18:2 to AA and DPA and 18:3 to EPA and DHA require **elongation and desaturation**.

The starting material (20:3, 20:4, 20:5) determines the number and placement of double bonds and, therefore, the number in the $\text{PGE}_\#$

Biosynthesis will form a cyclopentane ring and then modifying the resulting ring at the positions indicated in red.

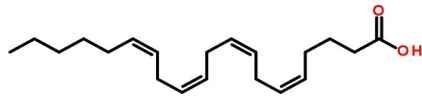


Outline

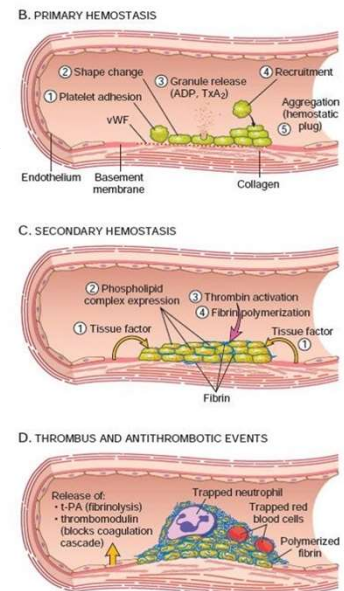
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Eicosanoids

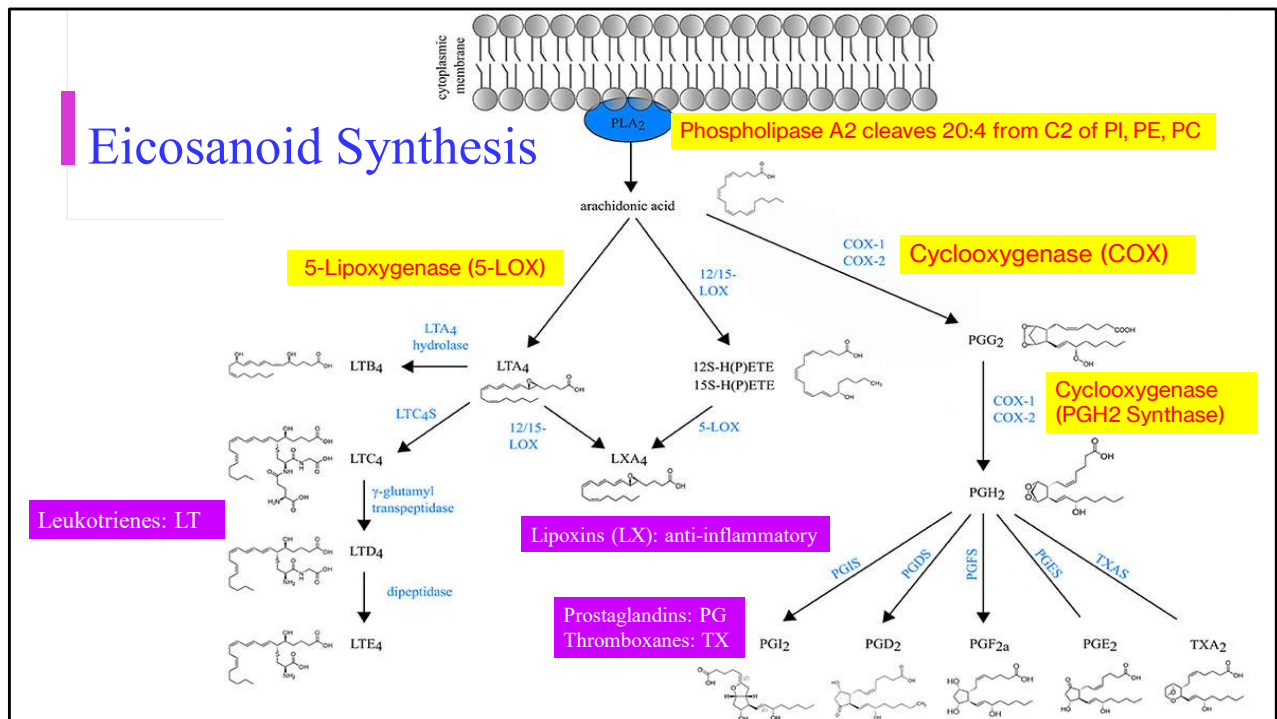
- Eicosanoids (“20-carbon” molecules) are derived from phospholipids.
- The ω -6 derived eicosanoids are generally:
 - pro-inflammatory molecules; short half-lives
 - act locally to initiate and promote inflammation and/or platelet aggregation.
- They also promote the innate immune response and coagulation (formation of blood clot – thrombus)



Arachidonic acid: 20:4



From Robbins Pathology p. 83 The innate immune response consists of physical, chemical and cellular defenses against pathogens. The main purpose of the innate immune response is to immediately prevent the spread and movement of foreign pathogens throughout the body.



Schematic overview of eicosanoid biosynthesis:

Arachidonic acid released from membrane phospholipids by cytosolic phospholipase A2 can be enzymatically converted either to prostaglandins (PGs) and thromboxane (TXA₂) by COX enzymes or to leukotrienes (LTs) and lipoxins (LXA₄s) by lipoxygenases (LOXs).

5-LOX: 5-lipoxygenase

12/15-LOX: 12/15-lipoxygenase

LTC₄S: LTC₄ synthase

PGIS: PGI or prostacyclin synthase

PGDS: PGD₂ synthase

PGFS: PGF synthase

PGES: PGE synthase

TXAS: TXA₂ synthase

Precursor of eicosanoids: **Arachidonic acid** (20:4, eicosatetraenoic acid)

Precursor of arachidonic acid: Linoleic acid (18:2, omega-6 FA)

- Eicosanoids are typically stored within cells.
- Synthesis is on-demand and is affected by physical, chemical and hormone stimuli.

Eicosanoids:

Prostaglandins: PG

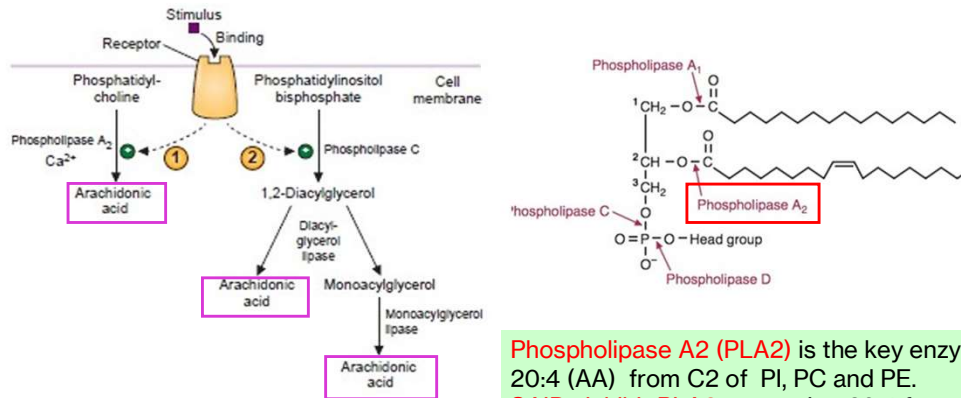
Thromboxanes:TX (e.g. **TXA2** –strong platelet aggregation)

Leukotrienes:

Critical enzymes:

1. PLA2- Phospholipase A2 releases arachidonic acid (20:4) from C2 of phospholipids like phosphatidyl inositol (PI) inhibited by SAIDs (steroidal anti-inflammatory drugs like cortisol and other glucocorticoids)
1. COX – cyclooxygenase – inhibited by NSAIDs (non-steroidal anti-inflammatory drugs, e.g. aspirin, ibuprofen)
2. LOX –lipoxygenase

Release of Arachidonic Acid from Phospholipids



Phospholipase A₂ (PLA₂) is the key enzyme in releasing 20:4 (AA) from C2 of PI, PC and PE.
 SAIDs inhibit PLA₂ preventing 20:4 from going into the eicosanoid pathway.
 Phospholipase C can also trigger the release of 20:4 from DAG.

Bacterial endotoxins stimulate increased biosynthesis of cytosolic **PLA2 protein** via a pathway involving mitogen-activated protein kinase.

Platelet activating factor acts through a 7-helix transmembrane receptor to stimulate **increased secretory PLA2 protein**. The binding of agonists such as thrombin, epinephrine, or collagen, to platelet surface receptors can trigger the **activation** of **phospholipase C** to catalyze the release of arachidonic acid from two major membrane phospholipids, **phosphatidylinositol and phosphatidylcholine**. Phospholipase C's role in signal transduction is its cleavage of phosphatidylinositol 4,5-bisphosphate (PIP₂) into diacyl glycerol (DAG) and inositol 1,4,5-trisphosphate (IP₃), which serve as second messengers.

Activators of each PLC vary, but typically include heterotrimeric G protein subunits, protein tyrosine kinases, small G proteins, Ca²⁺, and phospholipids.



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Cyclization of Arachidonic Acid

Cyclization of arachidonic acid will generate prostaglandins (PG) and thromboxanes (TX)

Key enzyme: **Cyclooxygenase (COX)**
Prostaglandin endoperoxidase synthase

Parent prostaglandin: **PGH₂**

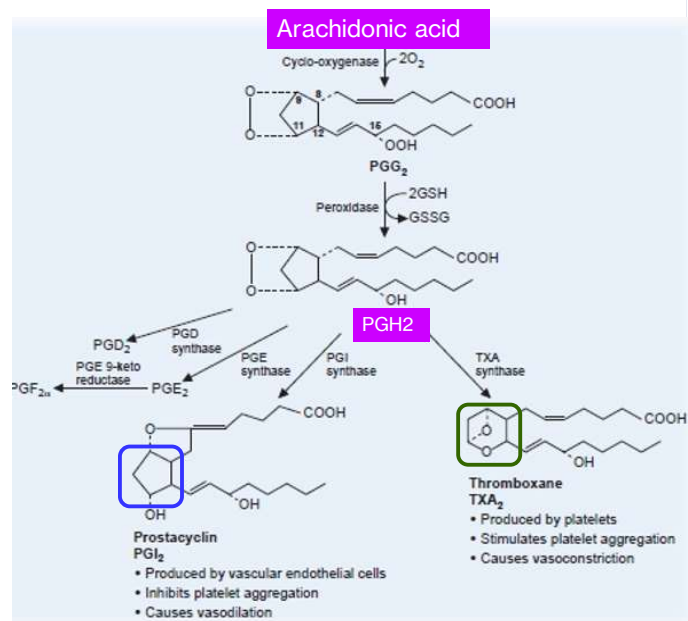
PGs have a **cyclopentane** ring

TXs have a **cyclooxane** ring

Nomenclature:

Third letter (A...K) – nature and position of substituents around the ring

Number – number of double bonds outside of the ring



5-Hydroxyeicosatetraenoic acid is an eicosanoid, i.e., a metabolite of arachidonic acid. It is produced by diverse cell types in humans and other animal species.

Arachidonic acid 5-hydroperoxide (**5-hydroperoxyeicosatetraenoic acid, 5-HPETE**) is an intermediate in the metabolism of arachidonic acid by the ALOX5 enzyme in humans or Alox5 enzyme in other mammals. Arachidonate 5-lipoxygenase, also known as ALOX5, 5-lipoxygenase, 5-LOX, or 5-LO, is a non-heme iron-containing enzyme.

Key Points:

Eicosanoids are products derived from phospholipids

Environmental stimuli or ligands activate the breakdown of specific lipids (C-2) to release arachidonates

Arachidonates & similar phospholipid-derived products become

eicosanoids – localized hormones

Prostaglandins (PG_)

Thromboxanes (TX_)

Leukotrienes (LT_n)

Other hydroxyeicosanoic acids

Prostaglandins

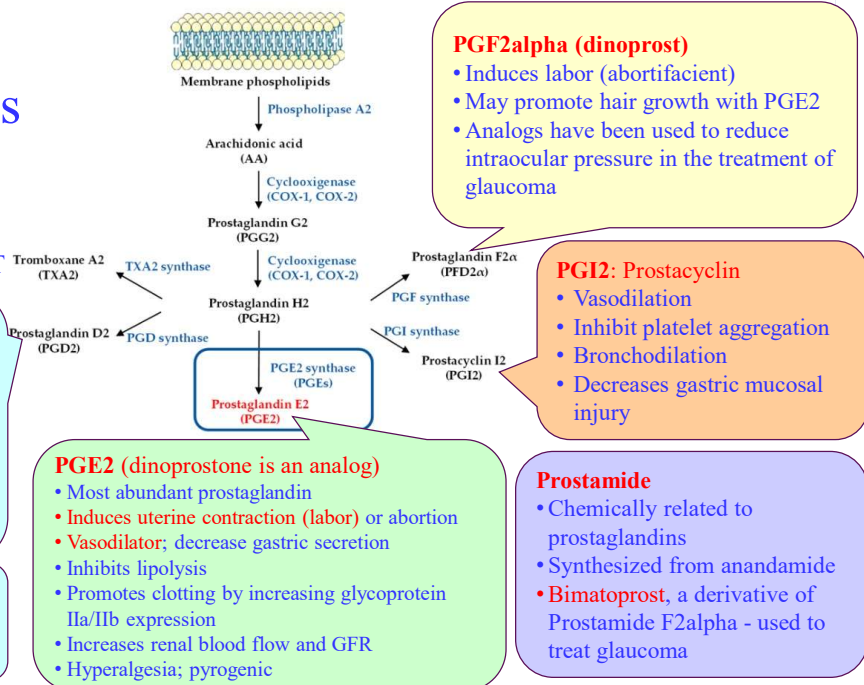
- Short half-lives and act locally.
- Increase secretion of protective mucus in the GIT
- Inhibit acid synthesis in the GIT

PGD2

- Asthma development
- Bronchoconstriction
- Cytokine signaling for T Helper2, eosinophils, basophils
- Inhibits hair growth
- Reduces body temperature during sleep

PGD2 → 15d-PGJ2

- 15d-PGJ2 is anti-inflammatory



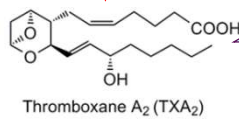
Prostaglandin E2 (**PGE2**), also known as dinoprostone, is a naturally occurring prostaglandin with oxytocic properties that is used as a medication. Dinoprostone is used in labor induction, bleeding after delivery, termination of pregnancy, and in newborn babies to keep the ductus arteriosus open.

PGD2 is the precursor of 15-deoxy- $\Delta^{12,14}$ -prostaglandin J2 (**15d-PGJ2**), a natural PPAR γ agonist, has been investigated for over a decade. Studies have revealed that it has proapoptotic, anti-inflammatory, antiangiogenic, and anti-metastatic abilities, as well as a significant anticancer effect. Activation of **PPAR- γ** causes insulin sensitization and enhances glucose metabolism. Peroxisome proliferator- activated receptor gamma (PPAR- γ)

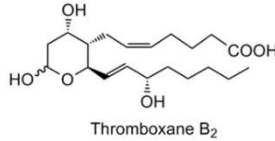
Thromboxanes

PGH₂

Thromboxane
synthase



nonenzymatic
hydrolysis



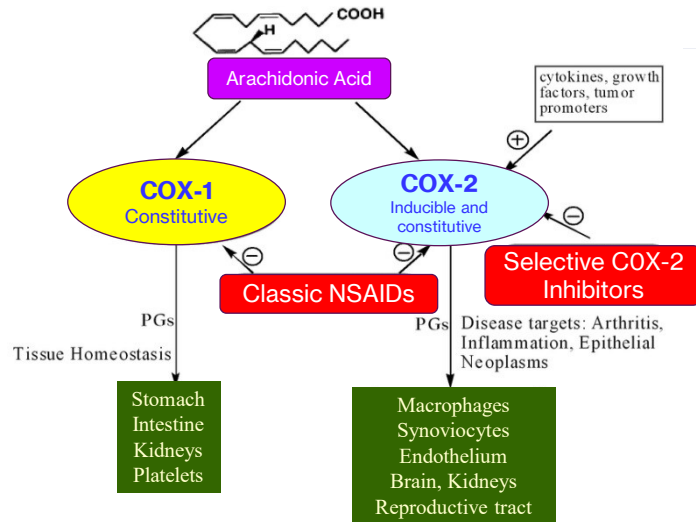
TXA₂

- Prothrombic
- Strong platelet aggregation
- Vasoconstriction
- Linked to tissue injury and inflammation
- Has increased activity in MI, stroke, atherosclerosis and bronchial asthma
- Also linked to pulmonary hypertension, kidney and hepatic injury, allergies, angiogenesis and cancer metastasis

TXB₂

- Inactive more stable metabolite of TXA₂
- Can be used to monitor response to aspirin therapy when used to prevent heart disease

Cyclo-oxygenase Isoforms



- **COX-1 is constitutive**, expressed in most tissues
 - Physiological and homeostatic role, cell signaling
 - Not influenced by steroid administration
 - Not increased by cytokines nor bacteria
- **COX-2 is inducible** but recently found to be constitutive in some tissues/organs
 - Found in immunocompetent cells (e.g., leukocytes)
 - Pathophysiological role, maintains inflammation
 - Induced by cytokines (interleukin-1), growth factors and tumor promoters
 - Inhibited by steroids



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Lipoxygenase Path

Leukotrienes (LTs)

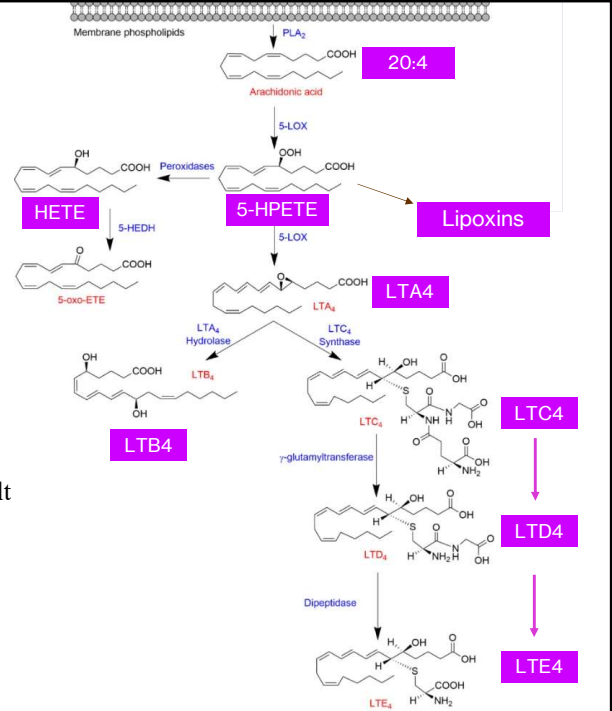
- First discovered in leukocytes
- Open structure with 3 conjugated double bonds
- **LTB₄**: chemotactic; involved in the recruitment and aggregation of neutrophils
- **LTC₄, LTD₄, LTE₄**: Slow reacting substances of anaphylaxis; bronchoconstriction; bronchospasm, increases vascular permeability

Lipoxins (LX)

- Derived from HPETE
- Generated in the resolution of acute inflammatory insult
- Anti-inflammatory

HETE (hydroxyeicosatetraenoic acid)

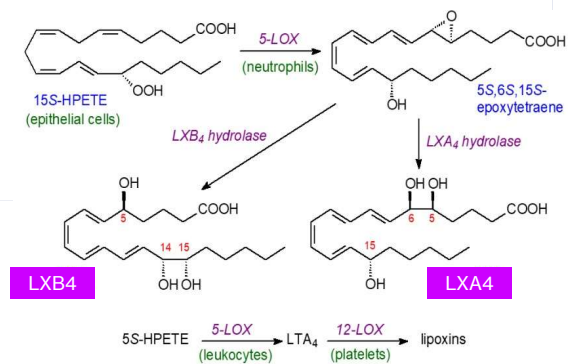
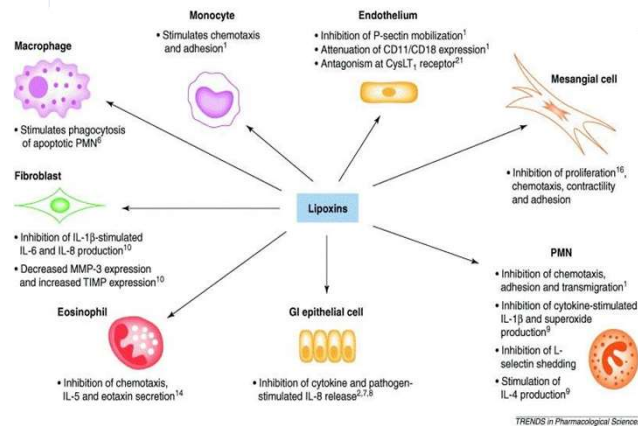
- Modulator of glomerular feedback in the kidneys
- Lung inflammatory mediator in lung disease; increase mucus, induces airway contraction



HETE = 5-Hydroxyeicosatetraenoic acid

HPETE = 5-hydroperoxyeicosatetraenoic acid, 5-HPETE

Lipoxins (LX)



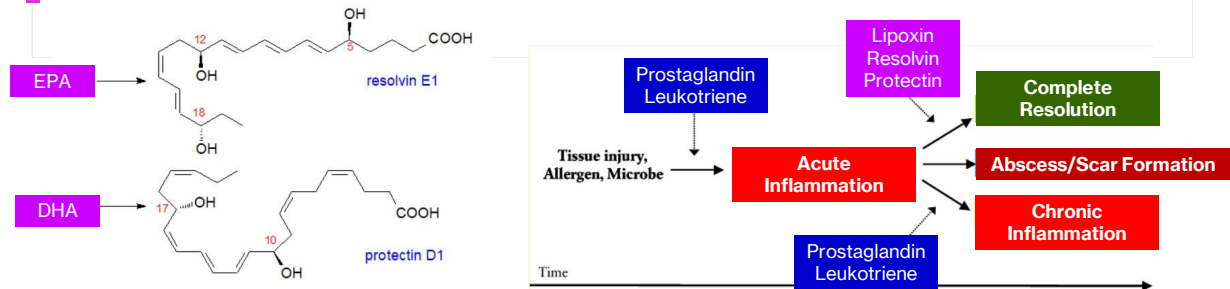
Lipoxins (LX) are arachidonate products formed by the sequential actions of the lipoxygenases (LOX). Lipoxins trigger the resolution or termination of inflammatory responses.

Lipoxins are ANTI- inflammatory – Low dosage of aspirin induces Aspirin Triggered (AT) Lipoxin production.

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Resolvins and Protectins from Omega-3 Fatty Acids



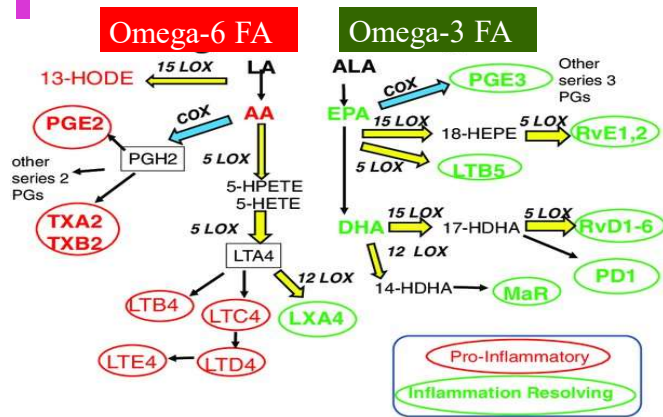
EPA (20:5) - Eicosapentaenoic acid
DHA (22:6) – Docosahexaenoic acid

Resolvins and protectins are anti-inflammatory. Aspirin can induce the synthesis of a family of resolvins from EPA.

Tissue injury activates the release of prostaglandins and leukotrienes from 20:4.

Lipoxins, resolvins and protectins are generated at a later time to initiate the resolution of inflammation

Omega-6 vs. Omega-3 Fatty Acids



Eicosanoids and related compounds derived from omega-6 FA are more potent mediators of thrombosis and inflammation than similar molecules derived from omega-3 FA (18:3, EPA, DHA)

↑↑Omega-3 FA in the diet:

↓↓PGE2, ↓↓TXA2, ↓↓LTB4

↑↑TXA3 – weak platelet aggregator

↑↑LTB5 – weak inducer of inflammation

↑↑Soybean oil
↑↑Sunflower oil
↑↑Corn Oil

ω-6 FA Diet ω-3 FA

Which diet do you follow?

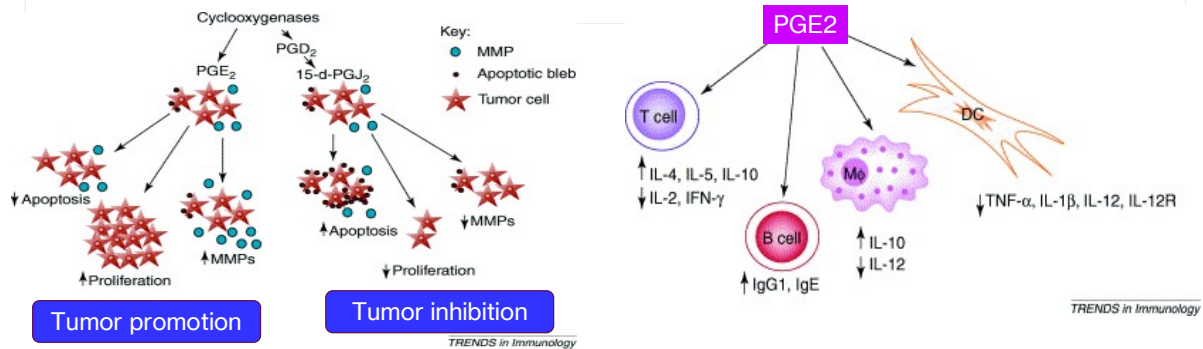
ω-6 FA Diet ω-3 FA

↑↑Fish Oil
↑↑Flax seed

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Prostaglandins in Infection, Cancer and Disease



PGE₂, PGD₂ and PGJ₂ mediate apoptosis and cell proliferation.

MMP- matrix metallo-proteinase

PGE₂ plays an important role in the development and regulation of the different immune cells. **PGE₂** has also been found to be elevated in periodontitis.

Matrix metalloproteinases (MMPs) are a family of zinc-containing endo-proteinases that share structural domains but differ in substrate specificity, cellular sources, and inducibility. MMPs of both T cells and macrophages facilitate secretion of TNF-alpha, by cleavage of the membrane- bound form.

Periodontitis is a chronic inflammatory disease involving interactions among bacterial products, host cells, and inflammatory mediators. The inflammatory response results in destruction of the tissue and alveolar bone supporting the teeth and can ultimately lead to tooth loss. The

inflammatory mediator prostaglandin E₂ (PGE₂) has been identified as a potent mediator in the pathogenesis of periodontitis. Levels of PGE₂ are elevated in the gingival tissue and gingival fluid of patients with periodontitis, compared with periodontally healthy subjects. It has also been reported that the inhibition of PGE₂ using selective or nonselective nonsteroidal anti-inflammatory drugs decreases periodontal disease progression and reduces alveolar bone resorption, which highlights the significance of PGE₂ in the pathogenesis of periodontal disease.

In addition to these findings, bone resorption in lipopolysaccharide-treated mice has

been shown to be dependent on PGE₂ synthesis, a finding that might be explained by the observation that PGE₂ stimulates the formation of osteoclasts

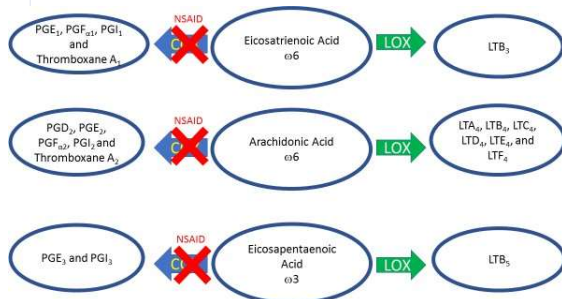
NSAIDs (Non-Steroidal Anti-Inflammatory Drugs)

- Reduce production of prostaglandins and thromboxanes
- Reduce pain, fever and inflammation
- First generation NSAIDs inhibit both COX-1 and COX-2 (*aspirin, ibuprofen, indomethacin, naproxen, phenylbutazone*)
 - Side effects include *gastrointestinal irritation, ulceration of stomach (H⁺), GIT bleeding which may lead to death if untreated (~16,000/yr in US)*
- COX-2 inhibitors(*Celebrex/Celecoxib*) are selective for COX-2
 - will not stimulate acid production in stomach
 - chronic usage caused heart valve problems
- Tylenol (*acetaminophen*) is thought to have effects through inhibiting the activity of COX-3, an alternatively spliced form of COX-1 present in the brain.

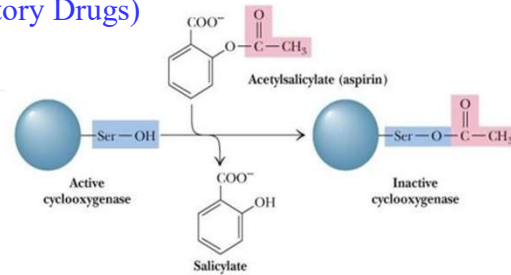
The **mechanism** of action of **celecoxib** is due to selective inhibition of cyclooxygenase-2 (COX-2), which is responsible for prostaglandin synthesis, an integral part of the pain and inflammation pathway. This pharmacologic activity gives **celecoxib** its analgesic, anti-inflammatory, and antipyretic effects.

The structures of several common NSAID's – anti-inflammatory & analgesic agents. Ibuprofen is sold as Motrin, Nuprin, and Advil. (b) Acetylsalicylate (aspirin) inhibits the COX activity of endoperoxide synthase *via* **acetylation (covalent modification) of Ser-530**. Acetaminophen (Tylenol) belongs to a class of drugs called analgesics.

NSAIDs (Non-Steroidal Anti-Inflammatory Drugs)



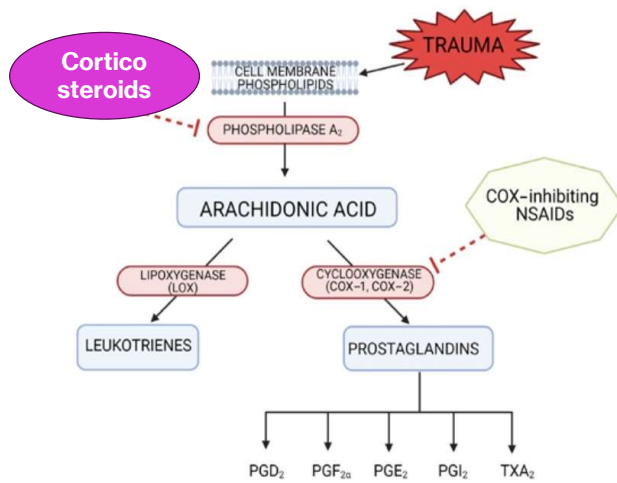
By inhibiting COX, NSAIDs can alter the activity of lipoxygenases (LOX) affecting the processing of 20:4 to HETE and leukotrienes.



LOW-DOSE ASPIRIN THERAPY (~75 mg/day)

- Aspirin acetylates the serine in the active site of COX-1 and COX-2 (IRREVERSIBLE INHIBITOR)
- Reduces TXA₂ preventing platelet aggregation
- Blood-thinner; reduces ability for blood to clot, thereby reducing heart attacks and stroke
- Side-effect: Bleeding especially of the GIT

SAIDs (Steroidal Anti-Inflammatory Drugs)



- **Corticosteroids (Glucocorticoids):**
Cortisol, prednisolone, betamethasone, dexamethasone, hydrocortisone
- **Inhibit Phospholipase A₂ (PLA₂)** preventing the release of arachidonic acid from C2 of phospholipids like PI, PC, PE
- Can also be used to treat asthma, allergies, rheumatoid arthritis, inflammatory bowel syndrome (IBS)
- Used as immunosuppressants in organ transplantation and autoimmune disorders
- Long term use may lead to **Iatrogenic Cushing Syndrome** and other complications

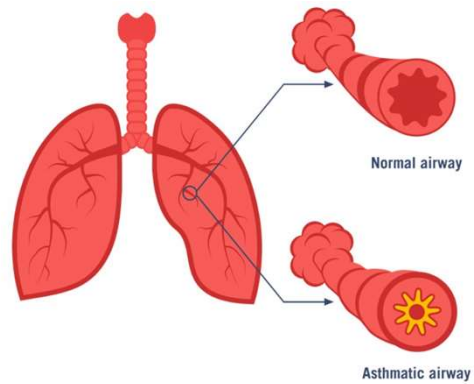
Corticosteroids and Leukotriene Antagonists for Asthma

- **Glucocorticoids**

- Beclomethasone (Beclovent, Vanceril)
- Fluticasone propionate (Flovent)

- **Alternatives to glucocorticoids:**

- **Leukotriene Receptor Antagonist**
 - Zafirlukast (Accolate)
 - Montelukast (Singulair)
- **Enzyme 5-Lipoxygenase Inhibitor**
 - Zileuton (Zyflo)



Summary

- Essential Fatty Acids are “essential” because they are required from the diet. (Humans lack a desaturase enzyme for the n-3 & n-6 positions in FA)
- ω -6 (n-6) Linoleic Acid is elongated and desaturated to produce Arachidonic Acid (Arachidonates)
- Arachidonates, through the actions of prostaglandin endoperoxidase synthase (COX), become autocrine and paracrine mediators of inflammation – prostaglandins & thromboxanes
- Aspirin, and other NSAID's, inhibit the cyclooxygenase enzyme activity within prostaglandin endoperoxidase synthase i.e., COX-1 & COX-2
- Additionally, low doses of aspirin have been demonstrated to produce AT-Lipoxin, anti-inflammatory compounds produced from arachidonates
- ω -3 (n-3) Linolenic Acid is elongated and desaturated to produce eicosapentanoic Acid (EPA) and docosahexaenoic Acid (DHA)
- EPA & DHA compete with eicosanoids through incorporation into phospholipids, produce anti-inflammatory prostaglandins & thromboxanes, as well as products that induce the resolution of inflammation

Study with all your might!

The key of persistence opens all doors closed by resistance.

John Di Lemme

Drop a message for questions and clarifications:

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Exploring Our Molecular Minds

Recall. Reflect. Validate.

Keyword Charts

Practice Makes Perfect

Keyword Chart 1: *Can you remember important info about the following keywords?*

Keyword	Description/property/clinical correlate
Arachidonic acid	
Cyclooxygenase	
LTE4	
Phospholipase A2	
TXA2	
Linolenic acid	

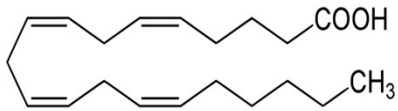
Keyword Chart 2

Identify the **Compound/Drug/Enzyme** described in the second column.

<i>Disorder</i>	<i>Clinical/Biochemical Info</i>
	Leukotriene implicated in chemotaxis
	Acetylates the serine in the active site of both Cox-1 and Cox-2
	Key enzyme in the synthesis of leukotrienes
	Class of compounds/drugs that inhibit Phospholipase A2; can be used as anti-inflammatory or anti-asthma drugs. Immunosuppressive
	Anti-inflammatory compound derived from EPA
	Precursor of arachidonic acid; essential FA

Practice Makes Perfect!

A 2-month-old infant has been found to have a defect in storing up the molecule shown below. Laboratory results will show low levels of which lipid?



- A. Linolenic acid
- B. Palmitic acid
- C. Phosphatidylinositol
- D. Cholesterol
- E. GM2

Practice Makes Perfect!

A 60-year-old woman has been advised by her physician to take 81 mg of aspirin per day to reduce the risk of blood clots leading to a heart attack. The rationale for this treatment is which of the following?

- A. To reduce LTC₄ synthesis
- B. To reduce TXA₂ synthesis
- C. To increase omega-3 synthesis
- D. To reduce PGD₂ synthesis
- E. To reduce cholesterol synthesis

Practice Makes Perfect!

A 28-year-old medical student was rushed to the QEH with rashes all over his body after eating a lobster at a local restaurant. He had felt very weak and had progressive breathing difficulty. The progressive breathing difficulty can be explained by which of the following processes?

- A. Hyperstimulation of cyclo-oxygenase
- B. Production of LTE₄
- C. Inhibition of cyclo-oxygenase
- D. Synthesis of linoleic acid
- E. Defect in porphyrin degradation